MIPT & YSDA · AUTUMN 2026

# Deep Generative Models

## Lecture 6

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# Recap of Previous Lecture

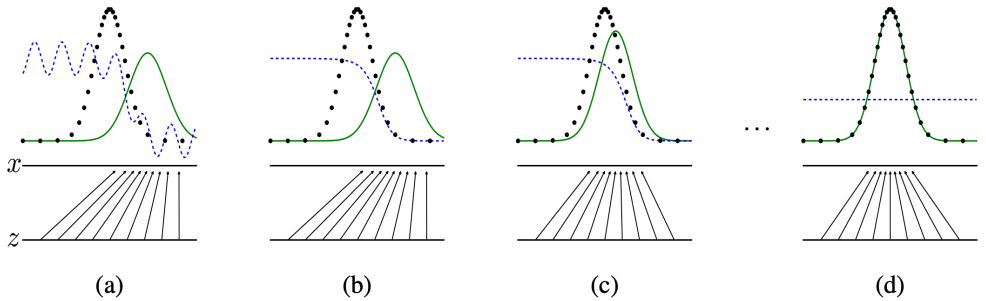
## Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$

## Optimal Discriminator and Jensen-Shannon Divergence

For fixed $G$, write the bracketed objective as $V(G, D)$:

$$D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}, \qquad V(G, D^*) = 2\, D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x})) - 2 \log 2.$$




Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014
Metz L. et al. Unrolled Generative Adversarial Networks, 2016

# Recap of Previous Lecture

GAN losses also train the reconstruction stage of a **latent generative model**.

## Stage 1: Train the Autoencoder

A fixed pretrained network $F$ defines the perceptual loss for reconstruction $\hat{\mathbf{x}}$:

$$\mathcal{L}_{\text{perc}} = \|F(\mathbf{x}) - F(\hat{\mathbf{x}})\|_2^2.$$

Train a discriminator $D$ to separate real and reconstructed image patches; train the autoencoder to fool $D$.

A KL-autoencoder also uses weak KL regularization.

## Stage 2: Freeze the Autoencoder and Learn a Prior

Fit the distribution of its latent codes — the **aggregated posterior**.

**VQGAN:** sample a code map $\mathbf{c}$ from an AR prior, look up its embeddings $\mathbf{z}_q(\mathbf{c})$, then decode.

$$\mathbf{c} \sim p_{\psi}(\mathbf{c}) \;\longrightarrow\; \mathbf{z}_q(\mathbf{c}) \;\longrightarrow\; \hat{\mathbf{x}}.$$

**KL-autoencoder:** a diffusion prior over continuous latents.

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \;\longrightarrow\; \mathbf{z} \sim p_{\psi}(\mathbf{z}) \;\longrightarrow\; \hat{\mathbf{x}}.$$

Esser P., Rombach R., Ommer B. Taming Transformers for High-Resolution Image Synthesis, 2021
Rombach R. et al. High-Resolution Image Synthesis with Latent Diffusion Models, 2022

# Recap of Previous Lecture

## Standard GAN Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$

## Theoretical Results

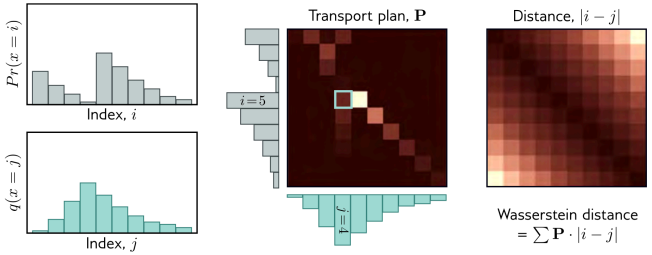
- Both the data distribution $p_{\text{data}}(\mathbf{x})$ and the generative distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ are supported on low-dimensional manifolds.
- If $p_{\text{data}}(\mathbf{x})$ and $p_{\boldsymbol{\theta}}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

$$D_{\text{KL}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = D_{\text{KL}}(p_{\boldsymbol{\theta}} \,\|\, p_{\text{data}}) = \infty, \qquad D_{\text{JS}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = \log 2$$

Weng L. From GAN to WGAN, 2019
Arjovsky M., Bottou L. Towards Principled Methods for Training Generative Adversarial Networks, 2017

# Recap of Previous Lecture




## Wasserstein Distance

$$W(\pi\|p) = \inf_{\gamma\in\Gamma(\pi,p)} \mathbb{E}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}\|\mathbf{x}_1 - \mathbf{x}_2\|$$

$$= \inf_{\gamma\in\Gamma(\pi,p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\|\gamma(\mathbf{x}_1,\mathbf{x}_2)d\mathbf{x}_1 d\mathbf{x}_2$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of "dirt" assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.
- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals $\pi$ and $p$ ($\int \gamma(\mathbf{x}_1, \mathbf{x}_2)d\mathbf{x}_1 = p(\mathbf{x}_2)$, $\int \gamma(\mathbf{x}_1, \mathbf{x}_2)d\mathbf{x}_2 = \pi(\mathbf{x}_1)$).
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Simon J.D. Prince. Understanding Deep Learning, 2023
Arjovsky M., Chintala S., Bottou L. Wasserstein GAN, 2017

# Recap of Previous Lecture

On a compact data space $\mathcal{X}$:

## Dependence on Generator Parameters

For a feedforward generator $\mathbf{G}_{\boldsymbol{\theta}}$ with Lipschitz activations and $\mathbb{E}_{p(\mathbf{z})}\|\mathbf{z}\| < \infty$, $W(p_{\text{data}}\|p_{\boldsymbol{\theta}})$ is continuous in $\boldsymbol{\theta}$ and differentiable almost everywhere.

## Convergence of Distributions

For a fixed distribution $\pi$ and a sequence $p_t$ on $\mathcal{X}$, as $t \to \infty$:

$$D_{\text{KL}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad D_{\text{JS}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad W(\pi\|p_t) \to 0$$

The first implication also holds for $D_{\text{KL}}(p_t\|\pi) \to 0$; the converses need not hold.

> **Wasserstein can capture distributions getting closer even when their supports do not overlap.**

# Recap of Previous Lecture

### Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right],$$

where $f : \mathbb{R}^m \to \mathbb{R}$ is $K$-Lipschitz continuous ($\|f\|_L \leq K$).

### WGAN Objective

$$\min_{\boldsymbol{\theta}} W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) \approx \min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})) \right].$$

- The discriminator $D$ is replaced by function $f$: in WGAN, it is known as the **critic**, which is *not* a classifier.
- If the weights $\boldsymbol{\phi}$ are restricted to a compact set $\boldsymbol{\Phi} = [-c, c]^d$, then $f_{\boldsymbol{\phi}}(\mathbf{x})$ is $K$-Lipschitz continuous.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) &= \max_{\|f\|_L \leq K} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right] \\ &\geq \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) \right] \end{aligned}$$

# Recap of Previous Lecture

## Frechet Inception Distance (FID)

Use Inception image embeddings $\mathbf{z} = \mathbf{f}(\mathbf{x})$; compare Gaussian approximations of their distributions:

$$\begin{aligned}\mathrm{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\boldsymbol{\theta}}\|^2 \\ + \mathrm{tr}\left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}} - 2\left(\boldsymbol{\Sigma}_{\text{data}}^{1/2}\boldsymbol{\Sigma}_{\boldsymbol{\theta}}\boldsymbol{\Sigma}_{\text{data}}^{1/2}\right)^{1/2}\right].\end{aligned}$$

FID depends on the feature network, Gaussian approximation and sample size; it may not correlate with human evaluation.

## CMMD: Comparing CLIP Image Embeddings

For CLIP image-embedding distributions $P, Q$, use a Gaussian kernel $k$:

$$\mathrm{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

Here $\mathbf{z}, \mathbf{z}' \sim P$ and $\mathbf{w}, \mathbf{w}' \sim Q$ are mutually independent. This compares more than means and covariances.

Heusel M. et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium, 2017
Jayasumana S. et al. Rethinking FID: Towards a Better Evaluation Metric for Image Generation, 2024

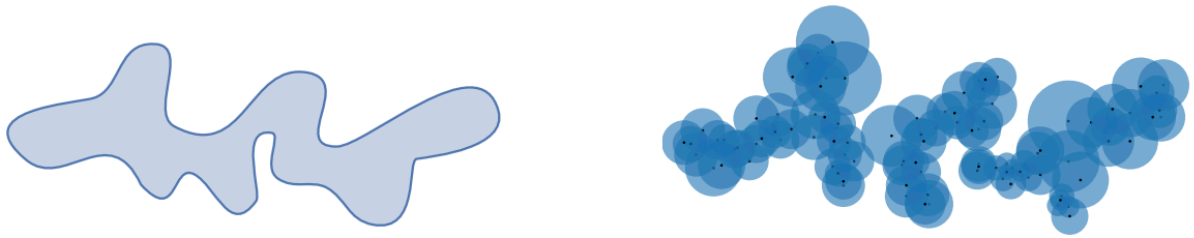
# Recap of Previous Lecture

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ - real samples;
- $\mathcal{S}_{\boldsymbol{\theta}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\boldsymbol{\theta}}(\mathbf{x})$ - generated samples.

Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists\, \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\boldsymbol{\theta}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\boldsymbol{\theta}}).$$



(a) True manifold (b) Approx. manifold

Embed the samples using a pretrained network (as in FID).

Kynkäänniemi T. et al. Improved precision and recall metric for assessing generative models, 2019

# Recap of Previous Lecture

Score an image $\mathbf{x}$ for a text prompt $t$.

## CLIP: Learn from Image–Caption Pairs

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)) \,.$$

## PickScore: Learn from Human Choices

For the same prompt, a human prefers $\mathbf{x}^+$ to $\mathbf{x}^-$:

$$\mathbb{P}_{\boldsymbol{\theta}}(\mathbf{x}^+ \succ \mathbf{x}^- \mid t) = \frac{\exp r_{\boldsymbol{\theta}}(\mathbf{x}^+, t)}{\exp r_{\boldsymbol{\theta}}(\mathbf{x}^+, t) + \exp r_{\boldsymbol{\theta}}(\mathbf{x}^-, t)}.$$

Train the score $r_{\boldsymbol{\theta}}$ to assign higher probability to the human choice.

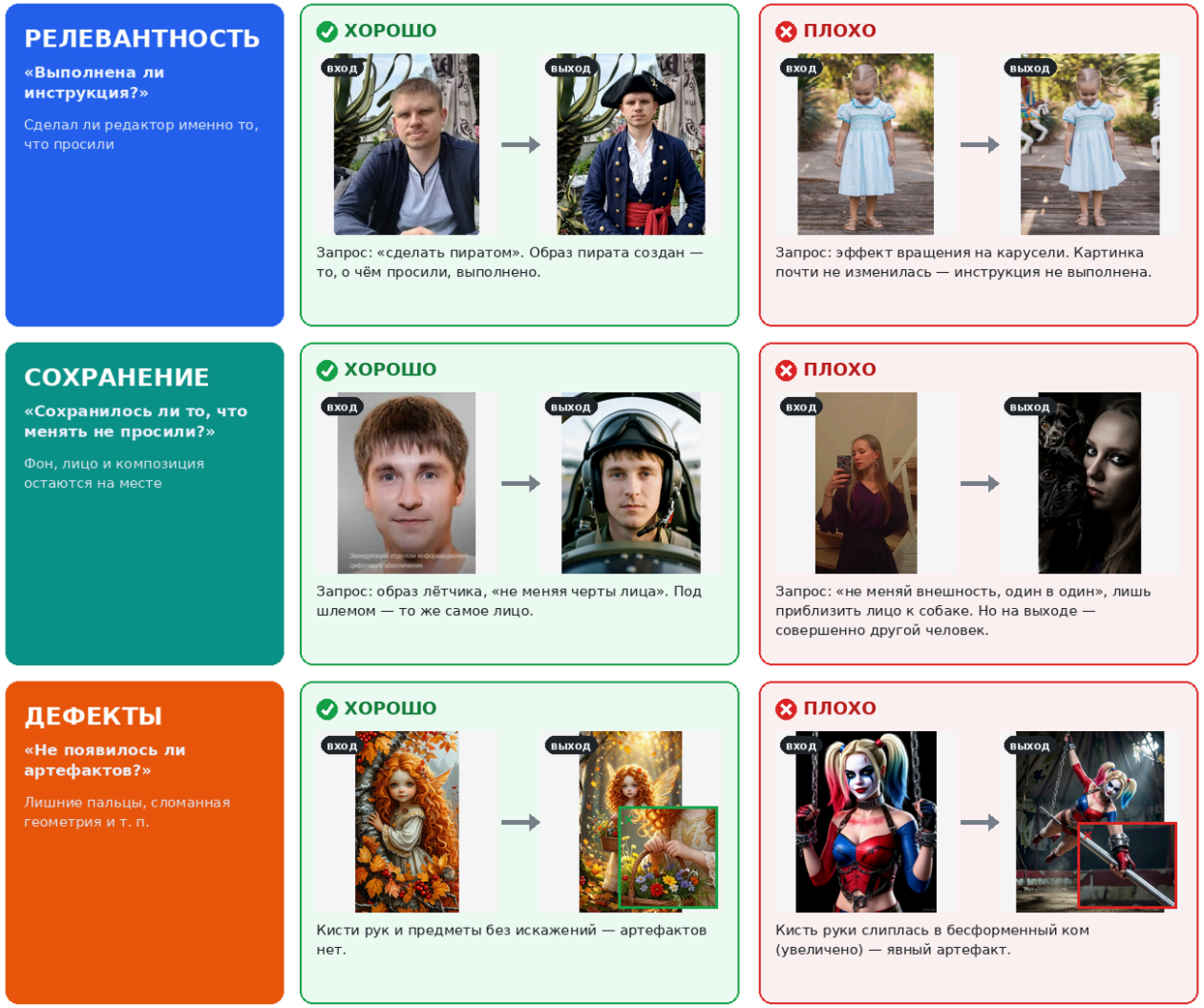
**VQAScore / GenEval / DPG-Bench:** evaluate compositional correctness; CLIP similarity can miss counts, attributes and relations.

A learned preference score can guide sample selection and model training — but optimizing the score can exploit its flaws.

Radford A. et al. Learning Transferable Visual Models From Natural Language Supervision, 2021
Kirstain Y. et al. Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation, 2023
Lin Z. et al. Evaluating Text-to-Visual Generation with Image-to-Text Generation, 2024
Ghosh D. et al. GenEval, 2023 · Hu X. et al. ELLA (DPG-Bench), 2024

# Recap of Previous Lecture

- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It’s important to assess various properties.



# Outline

01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

# Outline

01 **Langevin Dynamics**

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# Energy-Based Models

## Unnormalized Density

$$p_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})}{Z_{\boldsymbol{\theta}}}, \qquad \text{where } Z_{\boldsymbol{\theta}} = \int \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) d\mathbf{x}$$

- $\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})$ can be any non-negative function.
- If we reparameterize as $\hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) = \exp(-f_{\boldsymbol{\theta}}(\mathbf{x}))$, we eliminate the non-negativity constraint.

## Log-Density Gradient

The gradient of the normalized log-density equals that of the unnormalized log-density:

$$\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log Z_{\boldsymbol{\theta}} = \nabla_{\mathbf{x}} \log \hat{p}_{\boldsymbol{\theta}}(\mathbf{x})$$

- Suppose we already have this density (normalized or not) $p_{\boldsymbol{\theta}}(\mathbf{x})$.
- How can we sample from the model?

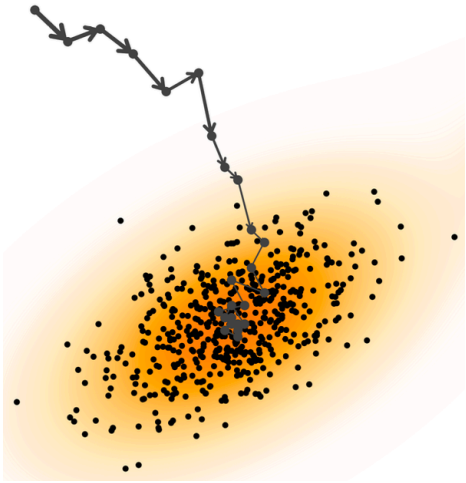
# Langevin Dynamics

## Theorem (Informal)

Let $\mathbf{x}_0$ be a random vector. Under mild regularity conditions, samples from the following dynamics will eventually follow $p_{\boldsymbol{\theta}}(\mathbf{x})$ (for sufficiently small $\eta$ and large $l$):

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

- What if $\boldsymbol{\epsilon}_l = \mathbf{0}$?
- The density $p_{\boldsymbol{\theta}}(\mathbf{x})$ is the **stationary** distribution of the Markov chain.
- The gradient is taken with respect to $\mathbf{x}$, not $\boldsymbol{\theta}$.
- $\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$ defines a vector field.

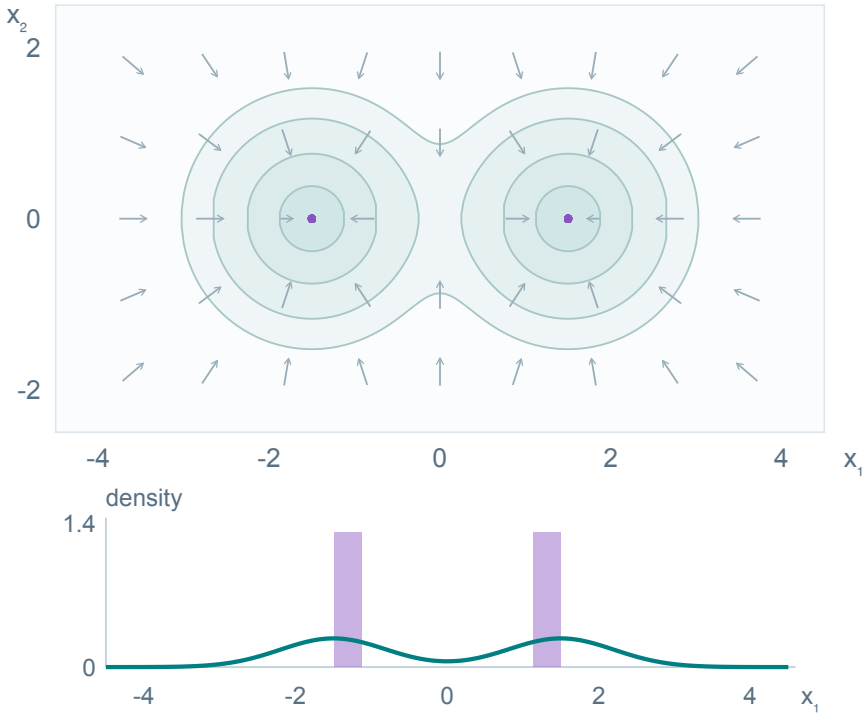


Lai C. H. et al. The principles of diffusion models, 2025.

# Langevin Dynamics: Why Add Noise?

$$p_{\theta}(\mathbf{x}) = \tfrac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{-}, 0.7^2\mathbf{I}) + \tfrac{1}{2}\mathcal{N}(\boldsymbol{\mu}_{+}, 0.7^2\mathbf{I}), \qquad \boldsymbol{\mu}_{\pm} = (\pm 1.5, 0), \quad \eta = 0.04.$$

**Noise off: approach the modes** Step 300




Density axis adapts to the histogram.

**Noise on: explore the density** Step 300




Density axis adapts to the histogram.

Finite step size and finite simulation time give approximate sampling.

# Outline

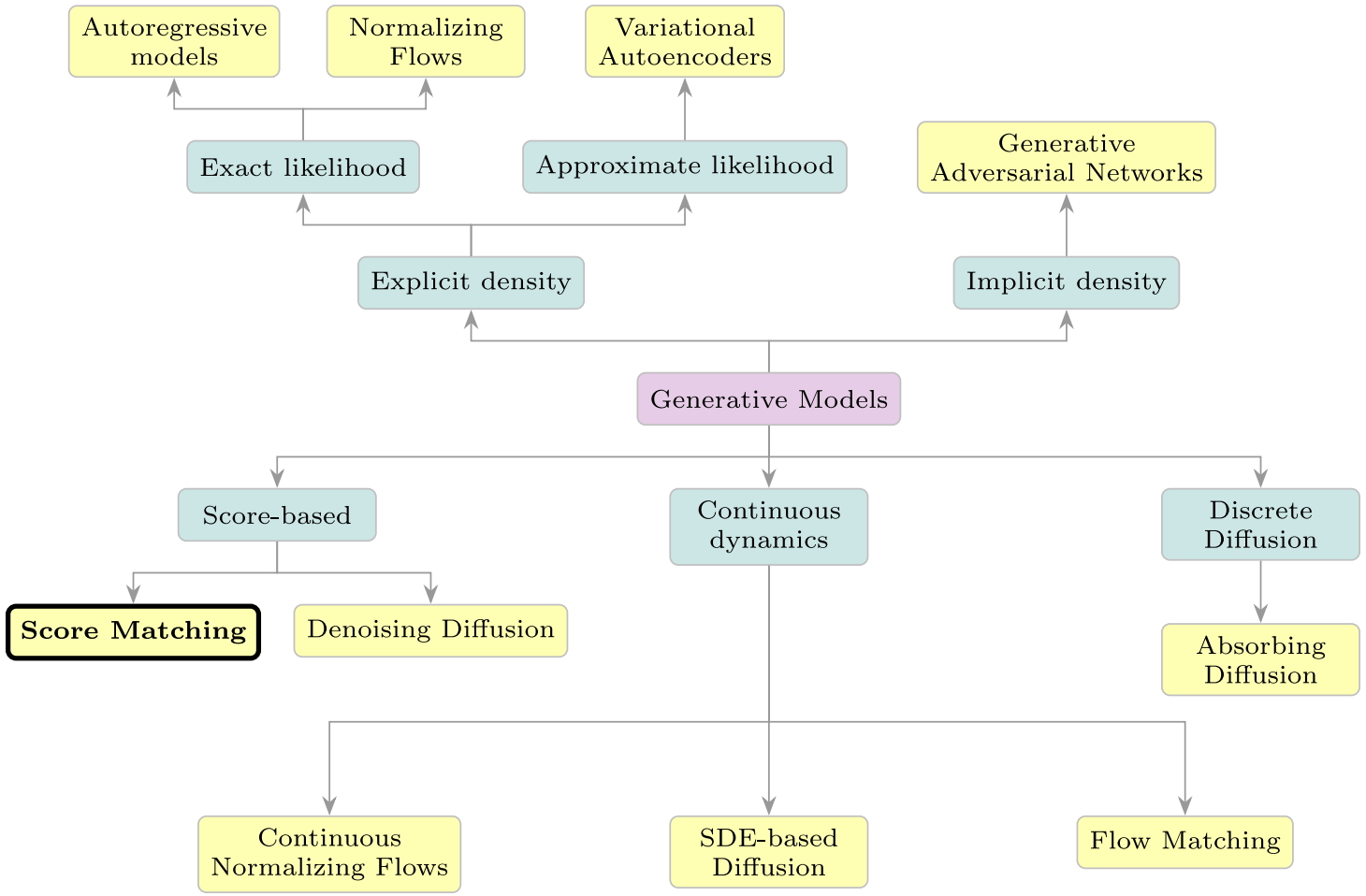
01 Langevin Dynamics

**02 Score Matching**

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

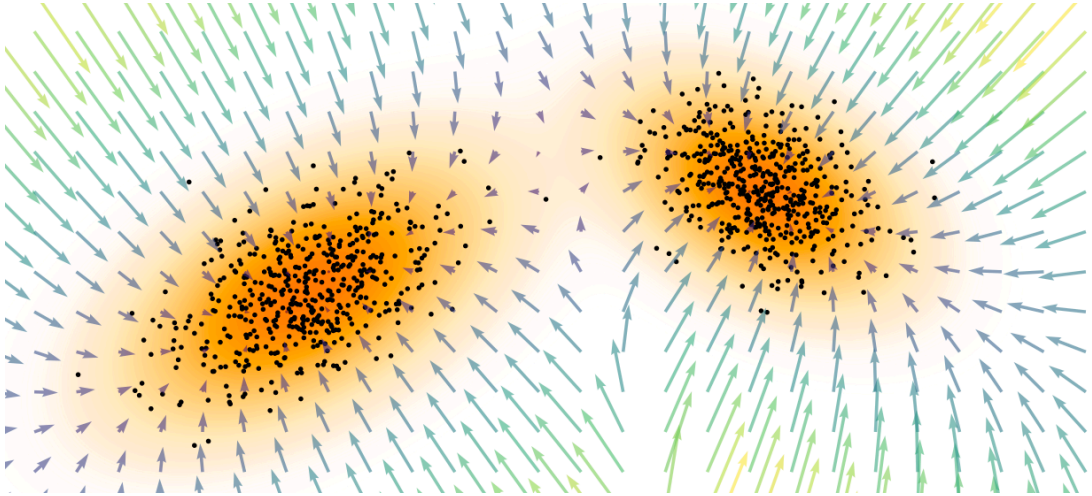
03 Forward Gaussian Diffusion Process

# Generative Models Taxonomy



# Score Function

$$\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x})$$

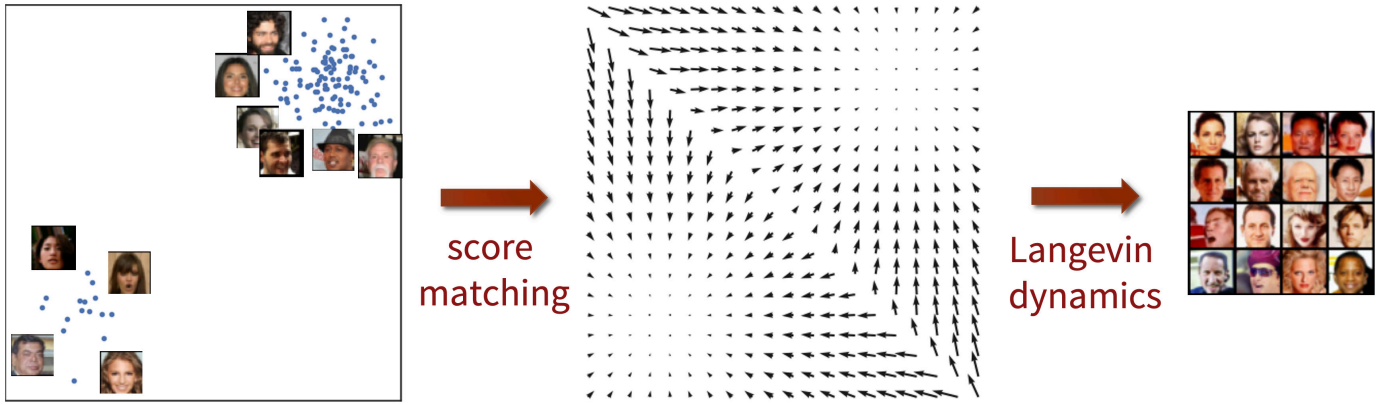


## Langevin Dynamics

$$\begin{aligned}
\mathbf{x}_{l+1} &= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\text{data}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l \\
&= \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l \\
&= \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l.
\end{aligned}$$

Lai C. H. et al. The principles of diffusion models, 2025.

# Score Matching




$$\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N\} \overset{\text{i.i.d.}}{\sim} p(\mathbf{x}) \qquad \mathbf{s}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

## Fisher Divergence

$$\begin{aligned} D_F(p_{\text{data}}, p_{\boldsymbol{\theta}}) &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \\ &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}} \end{aligned}$$

**Problem:** We don't know $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.

# Why Is the Score Function Better than the Density?

## Freedom from Normalization Constants

Many distributions are defined only up to an unnormalized density $\hat{p}(\mathbf{x})$:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z}_{=0} = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x})$$

The score function bypasses the intractable constant $Z$ entirely!

## A Complete Representation

The score function fully characterizes the underlying distribution (up to a constant):

$$\log p(\mathbf{x}) = \log p(\mathbf{x}_0) + \int_0^1 \mathbf{s}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))^\top (\mathbf{x} - \mathbf{x}_0) dt$$

Modeling the score is as expressive as modeling $p(\mathbf{x})$ itself, while often more tractable for generative modeling.

# Outline

01 Langevin Dynamics

**02 Score Matching**

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

# Denoising Score Matching

Let us perturb the original data $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ with Gaussian noise:

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad q(\mathbf{x}_\sigma | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$

$$q(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma | \mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}.$$

**Assumption**

The solution to

$$\frac{1}{2} \mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

satisfies $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta}, 0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ if $\sigma$ is sufficiently small.

- The score function of the noised data nearly matches the score function of the original data.
- The score function $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma)$ is parameterized by $\sigma$.
- **Note:** We don't know $q(\mathbf{x}_\sigma)$, just as we don't know $p_{\text{data}}(\mathbf{x})$.

# Denoising Score Matching

## Theorem

Under mild regularity conditions, the following holds:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}(\boldsymbol{\theta})$$

## Gradient of the Noise Kernel

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \qquad q(\mathbf{x}_\sigma|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$
$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) = -\frac{\mathbf{x}_\sigma - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}$$

- The right-hand side doesn't require computing $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$ or even $\nabla_{\mathbf{x}_\sigma} \log p_{\text{data}}(\mathbf{x}_\sigma)$.
- $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)$ is trained to **denoise** the noised samples $\mathbf{x}_\sigma$.

# Denoising Score Matching

Initial objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noised objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma} \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

## Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

# Denoising Score Matching

## Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}(\boldsymbol{\theta}).$$

## Proof

For any integrable $h$, marginalize the joint distribution:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int \left( \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma).$$

Expand the squared error:

$$\begin{aligned} &\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \\ = \mathbb{E}_{q(\mathbf{x}_\sigma)} &\left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \right\|_2^2 + \underbrace{\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2}_{C_1} - 2\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[ \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right]. \end{aligned}$$

The score-norm term $C_1$ is independent of $\boldsymbol{\theta}$.

# Denoising Score Matching

## Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}(\boldsymbol{\theta}).$$

## Proof (Continued)

Rewrite the mixed term:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[ \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right] = \int \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma) d\mathbf{x}_\sigma$$

$$= \int \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \left( \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) d\mathbf{x}_\sigma = \iint p_{\text{data}}(\mathbf{x}) \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x}) \, d\mathbf{x} \, d\mathbf{x}_\sigma$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[ \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right].$$

Substitute and complete the square:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[ \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \right\|_2^2 - 2 \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right] + C_1$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}(\boldsymbol{\theta}).$$

# Denoising Score Matching

## Denoising Score Matching Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The optimal MSE predictor is the posterior mean of the regression target:

$$\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} \left[ \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right]$$

## Marginal Score via Conditional Scores

$$\begin{aligned}
\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) &= \frac{\nabla_{\mathbf{x}_\sigma} \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}}{q(\mathbf{x}_\sigma)} \\
&= \int \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \cdot \frac{q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x})}{q(\mathbf{x}_\sigma)} d\mathbf{x} \\
&= \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)} \left[ \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right]
\end{aligned}$$

**Therefore:** $\mathbf{s}_{\boldsymbol{\theta}^*,\sigma}(\mathbf{x}_\sigma) = \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$ - the optimal model learns the true **marginal** score!

# Tweedie's Formula

Let $\mathbf{x} \sim p_{\text{data}}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be independent, with $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$ and $\sigma > 0$.

Using the posterior-mean identity from the previous slide:

$$\begin{aligned}\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) &= \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\frac{\mathbf{x} - \mathbf{x}_\sigma}{\sigma^2}\right] \\ &= \frac{\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] - \mathbf{x}_\sigma}{\sigma^2}.\end{aligned}$$

## Tweedie's Formula

$$\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

## Noise Prediction

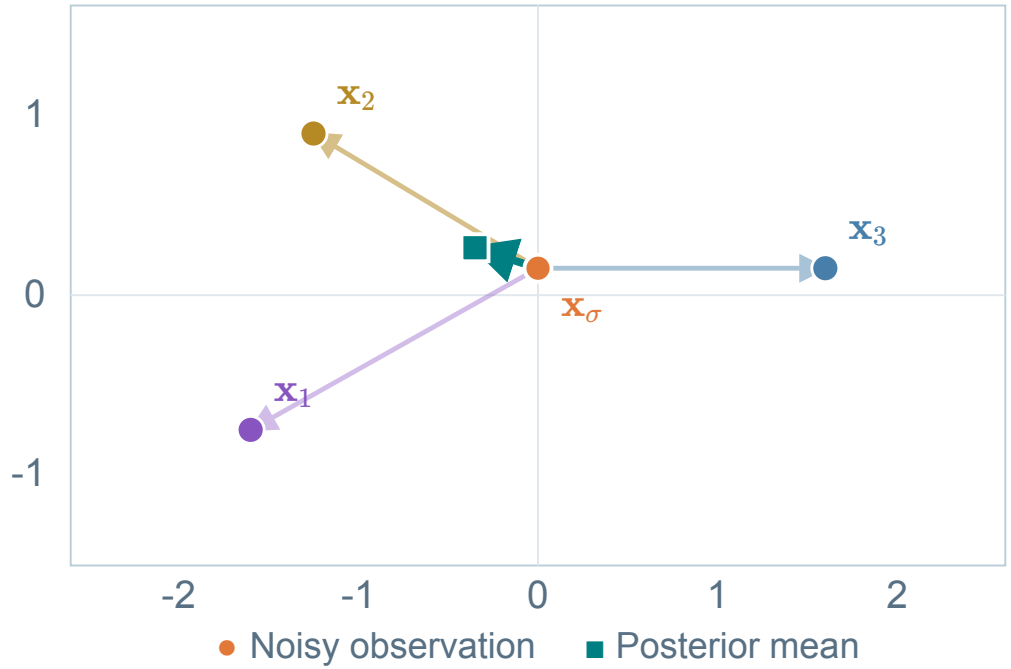
$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{x}_\sigma] = -\sigma \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

Both conditional means are MSE-optimal predictions given the noisy observation.

# From Conditional Scores to Denoising

Toy data: three equally likely clean points $\mathbf{x}_i$, with $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma = 1$.

Fixed noisy observation between the clean-data modes.




## Conditional scores

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}_i) = (\mathbf{x}_i - \mathbf{x}_\sigma)/\sigma^2$$

## Posterior weights

| $q(\mathbf{x}_1\|\mathbf{x}_\sigma)$ | $q(\mathbf{x}_2\|\mathbf{x}_\sigma)$ | $q(\mathbf{x}_3\|\mathbf{x}_\sigma)$ |
|---|---|---|
| 22.9% | 42.7% | 34.4% |

## Marginal score = posterior-weighted average

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \sum_i q(\mathbf{x}_i|\mathbf{x}_\sigma) \tfrac{\mathbf{x}_i - \mathbf{x}_\sigma}{\sigma^2}$$

## Tweedie: from the score to the clean-data mean

$$\mathbb{E}[\mathbf{x}|\mathbf{x}_\sigma] = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$$

Posterior mean: (-0.35, 0.26)

> The MSE-optimal denoised prediction can lie between the clean-data modes.

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010
Efron B. Tweedie's Formula and Selection Bias, 2011

# Denoising Score Matching

Original objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Noisy objective:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

This is equivalent to a denoising task:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma\boldsymbol{\epsilon}) + \frac{\boldsymbol{\epsilon}}{\sigma} \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

## Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

# Outline

01 Langevin Dynamics

**02 Score Matching**

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

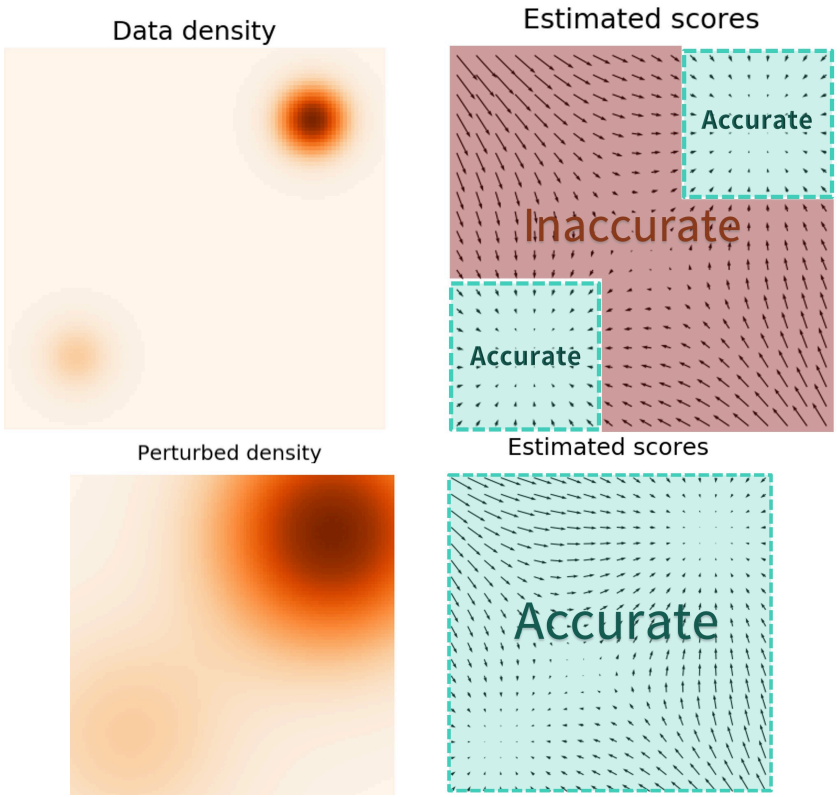
03 Forward Gaussian Diffusion Process

# Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathcal{N}(0,\mathbf{I})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}+\sigma\boldsymbol{\epsilon})+\frac{\boldsymbol{\epsilon}}{\sigma}\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2}\cdot\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta}\cdot\boldsymbol{\epsilon}_l$$

- For **small** $\sigma$, $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x})$ becomes inaccurate and Langevin dynamics fails to traverse modes.
- For **large** $\sigma$, robustness in low-density regions is achieved, but the model learns a distribution that is overly corrupted.

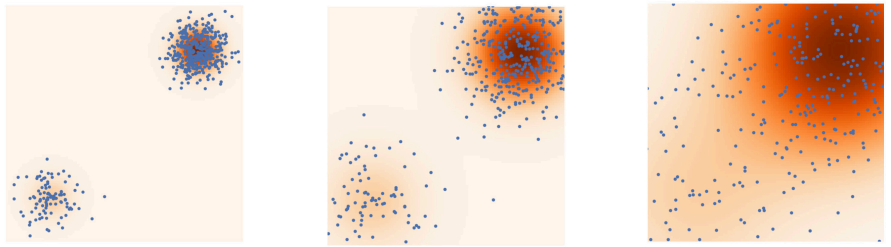


# Noise-Conditioned Score Network (NCSN)

- Specify a sequence of noise levels: $\sigma_1 < \sigma_2 < \cdots < \sigma_T$.
- Perturb each data point with different noise levels: $\mathbf{x}_t = \mathbf{x} + \sigma_t \boldsymbol{\epsilon}$, so $\mathbf{x}_t \sim q(\mathbf{x}_t)$.
- Choose $\sigma_1, \sigma_T$ such that:

$$q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \qquad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$$\sigma_1 < \sigma_2 < \sigma_3$$





# Noise-Conditioned Score Network (NCSN)

Train the denoising score function $\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t)$ for each noise level $\sigma_t$ using a unified weighted objective:

$$\sum_{t=1}^{T} \sigma_t^2 \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Here, $\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2} = -\frac{\boldsymbol{\epsilon}}{\sigma_t}$

## Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L} = \sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2$.
4. Update $\boldsymbol{\theta}$ via stochastic gradient descent.

How do we sample from such a model?

# Noise-Conditioned Score Network (NCSN)

## Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via $L$ steps of Langevin dynamics at each noise level $\sigma_t$:
$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2}\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t}\boldsymbol{\epsilon}_t^l.$$
3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to the next noise level $\sigma_{t-1}$.



# Outline

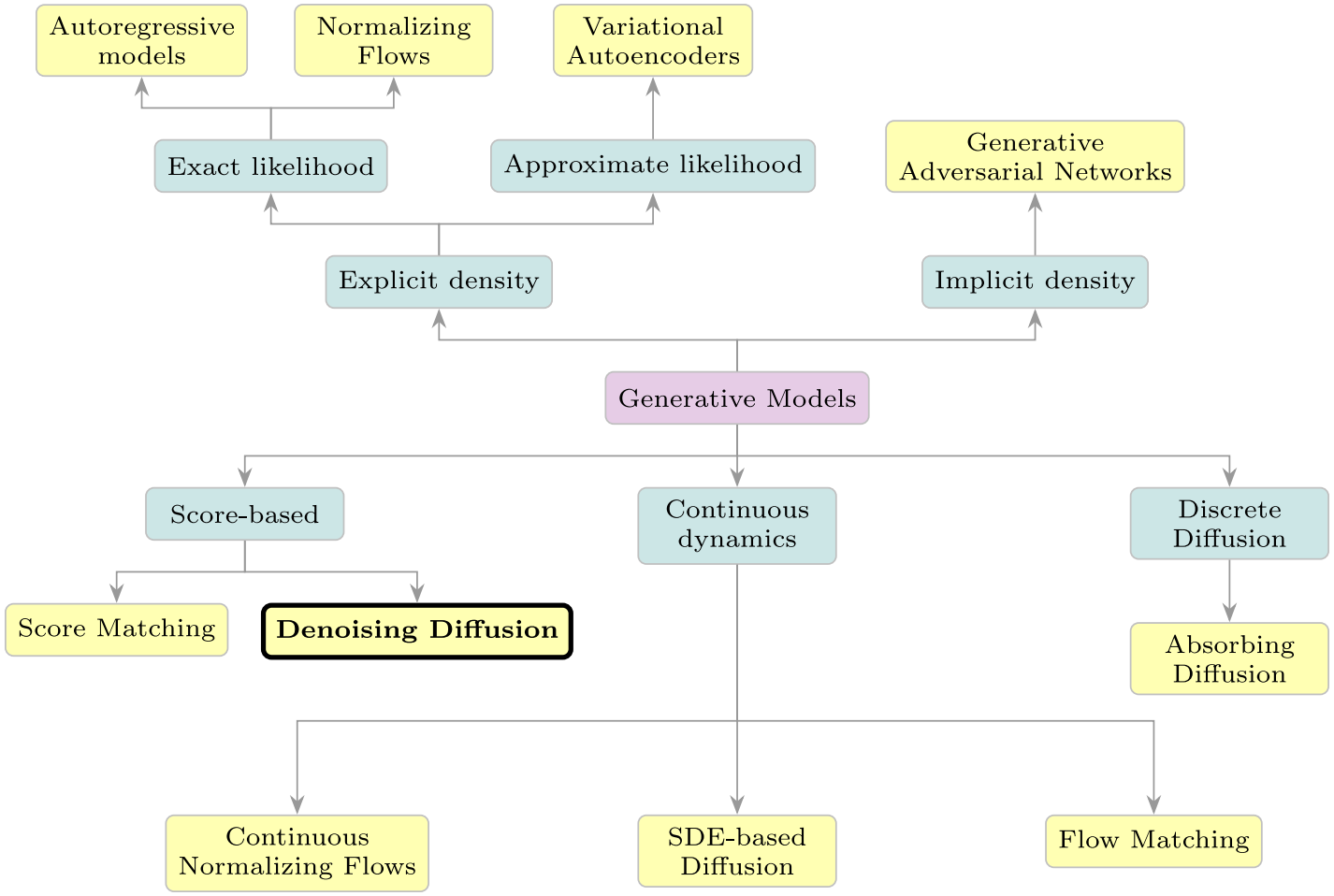
01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

**03 Forward Gaussian Diffusion Process**

# Generative Models Taxonomy



# Forward Gaussian Diffusion Process

Let $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\beta_t \ll 1$. Define a Markov chain:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

## Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&\approx \left(1 - \frac{\beta_t}{2}\right)\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&= \mathbf{x}_{t-1} + \frac{\beta_t}{2}(-\mathbf{x}_{t-1}) + \sqrt{\beta_t}\boldsymbol{\epsilon}_t
\end{aligned}$$

- $\beta_t = \eta$
- $\nabla_{\mathbf{x}_{t-1}} \log p_{\boldsymbol{\theta}}(\mathbf{x}_{t-1}) = -\mathbf{x}_{t-1} = \nabla_{\mathbf{x}_{t-1}} \log \mathcal{N}(0, \mathbf{I})$

# Forward Gaussian Diffusion Process

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

## Statement 1

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s = \prod_{s=1}^{t}(1-\beta_s)$. Then

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$

Thus, samples at any timestep $t$ can be generated directly from $\mathbf{x}_0$

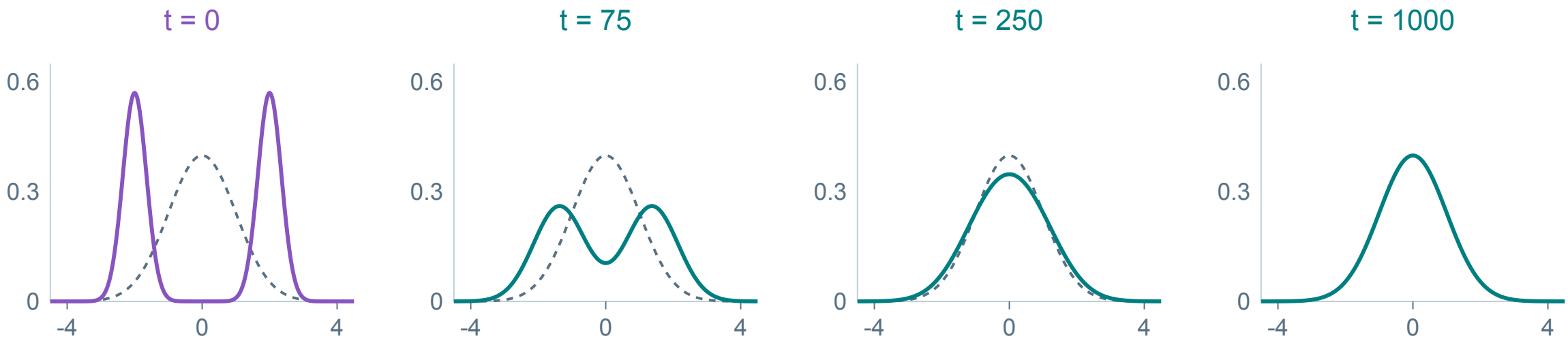
$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t}(\sqrt{\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_{t-1}}\boldsymbol{\epsilon}_{t-1}) + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + (\sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{\epsilon}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t) \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}'_t \\
&= \ldots = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})
\end{aligned}$$

# Forward Gaussian Diffusion Process

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right);$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$p_{\text{data}}(x_0) = \tfrac{1}{2}\mathcal{N}(-2, 0.35^2) + \tfrac{1}{2}\mathcal{N}(2, 0.35^2), \qquad \beta_t = 0.01$$




---- Standard Gaussian reference $\mathcal{N}(0, 1)$ .

The plots show marginal densities; the kernels above are conditional.

# Forward Gaussian Diffusion Process

## Statement 2

Applying the Markov chain to any distribution $p_{\text{data}}(\mathbf{x})$ yields $\mathbf{x}_\infty \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$, the **stationary** (limiting) distribution:

$$p_\infty(\mathbf{x}) = \int q(\mathbf{x}|\mathbf{x}')p_\infty(\mathbf{x}')d\mathbf{x}'$$

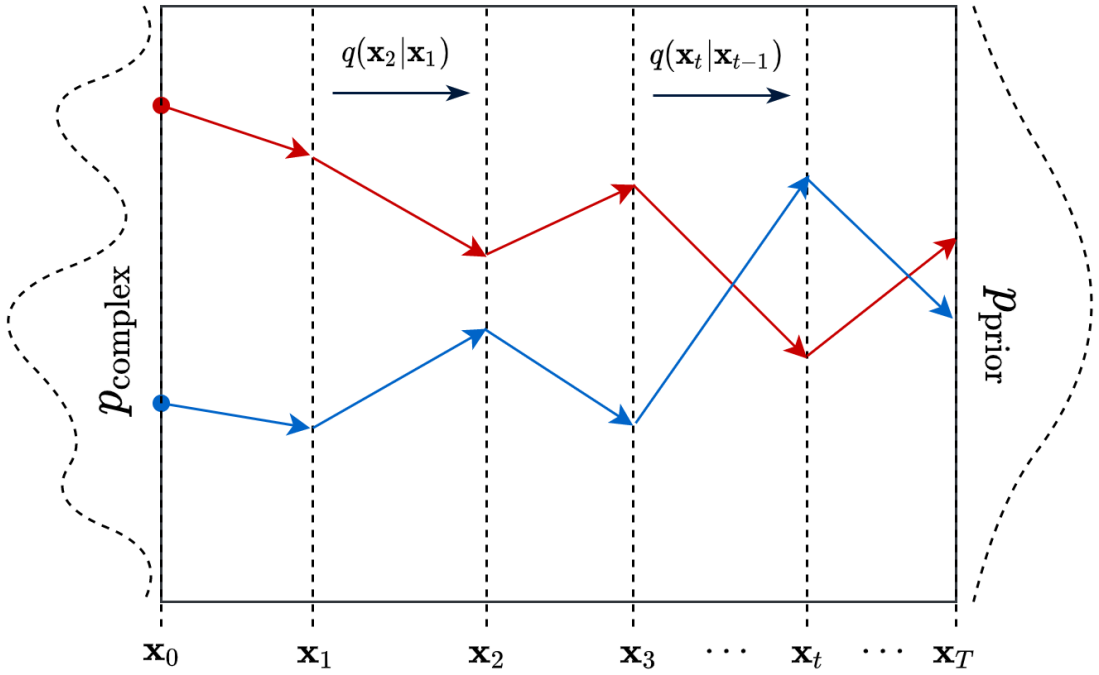
$$p_\infty(\mathbf{x}) = \int q(\mathbf{x}_\infty|\mathbf{x}_0)p_{\text{data}}(\mathbf{x}_0)d\mathbf{x}_0$$

$$\approx \mathcal{N}(0, \mathbf{I}) \int p_{\text{data}}(\mathbf{x}_0)d\mathbf{x}_0 = \mathcal{N}(0, \mathbf{I})$$

**Note:** This holds iff $\bar{\alpha}_t \to 0$, i.e., $\sum_{t=1}^{\infty} \beta_t = +\infty$.

# Forward Gaussian Diffusion Process

**Diffusion** describes the migration of particles from regions of high density to those of low density.




1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t,$
   $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}),\ t \geq 1$
3. After $T \gg 1$ steps: $\mathbf{x}_T \sim p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$

If this process can be reversed, we can sample from $p_{\text{data}}(\mathbf{x})$ by starting from noise $p_\infty(\mathbf{x}) = \mathcal{N}(0, \mathbf{I})$.
Our goal now becomes inverting this diffusion.

# Denoising Score Matching

## NCSN

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2\mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2\mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}$$

## Gaussian Diffusion

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1-\bar{\alpha}_t}$$

## Theorem (Denoising Score Matching)

$$\begin{aligned}&\mathbb{E}_{q(\mathbf{x}_t)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) \right\|_2^2 \\ &\quad = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 + \text{const}(\boldsymbol{\theta})\end{aligned}$$

**Note:** Annealed Langevin dynamics applies to diffusion, too.
NCSN with annealed Langevin dynamics is an early score-based generative framework (2019–2020).

# Summary

- Langevin dynamics combines log-density gradients and Gaussian noise to sample approximately from an unnormalized model.
- Score matching minimizes Fisher divergence to estimate the score function.
- Denoising score matching makes the score objective estimable using corrupted samples and the known corruption kernel.
- Tweedie’s formula converts the score into MSE-optimal clean-data and noise predictions.
- Noise-Conditioned Score Networks learn scores at multiple noise levels and sample with annealed Langevin dynamics.
- The forward Gaussian diffusion process admits a closed-form conditional distribution for any timestep.
- With a suitable noise schedule, the forward process converges to standard Gaussian noise; denoising score matching also applies to these diffusion marginals.